

1.

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Devices using SMPS (Switched Mode Power Supply)

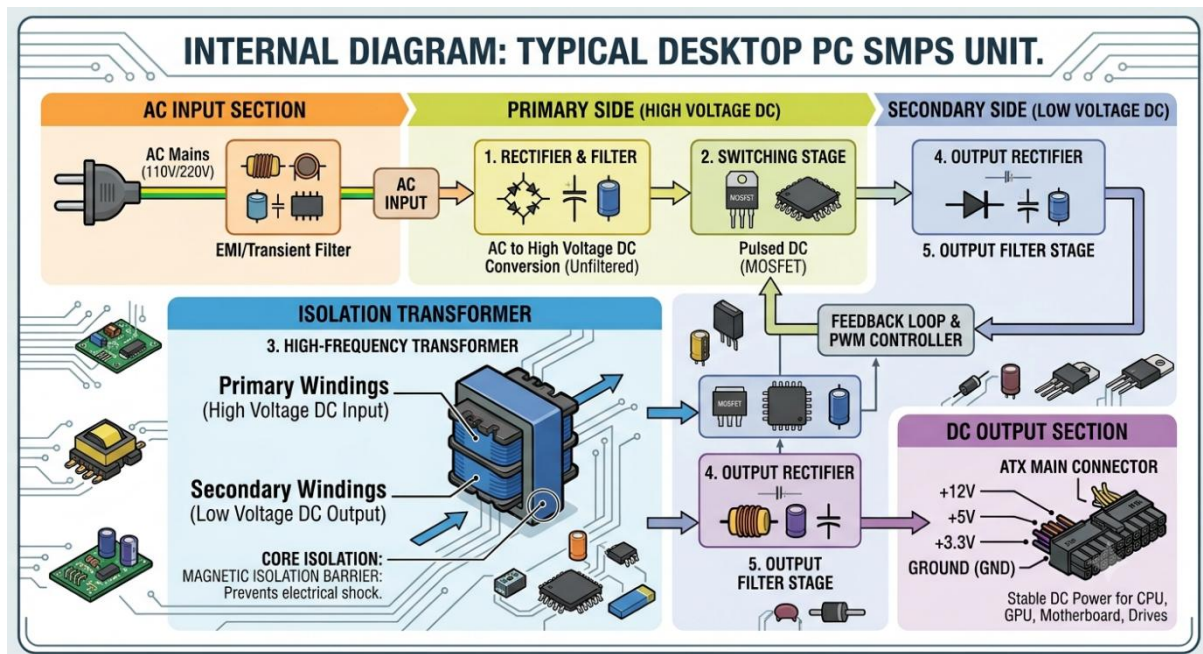
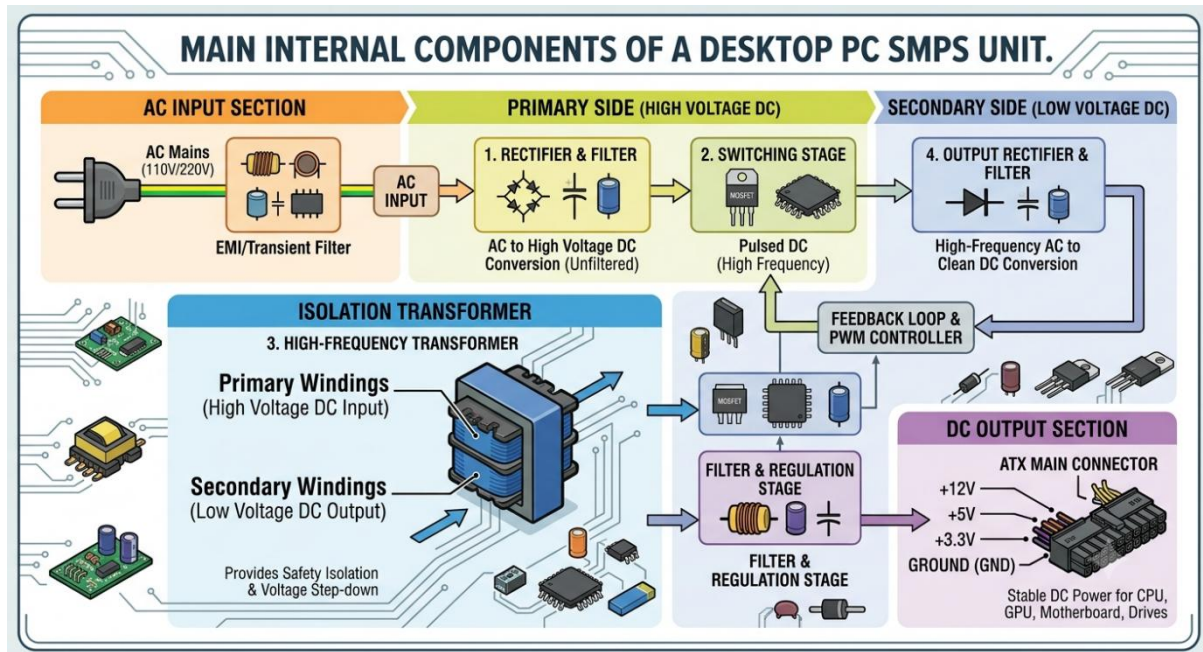
1. **Desktop Computer (PC Tower):** Safely converts electricity from your wall into the exact low-power amounts needed for the computer parts without overheating.
2. **Laptop Charger:** A small, lightweight box that turns wall power into the right power to charge your laptop battery safely.
3. **LED Smart TV:** Needs a tiny, efficient power system that fits inside a thin TV without taking up space or making it too heavy.

Devices using UPS (Uninterruptible Power Supply)

1. **Wi-Fi Router:** Keeps your internet running during a sudden power outage so your Wi-Fi doesn't turn off instantly.
2. **Desktop PC:** Gives you extra battery time during a power cut so you can save your work and turn off your computer safely.

2.

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



3.

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Motherboard Connector



ATX 24 pin motherboard power connector photo

- **Name:** ATX 24-Pin Power Connector
- **What it powers:** The motherboard (main circuit board), which passes power to the CPU, RAM, and small attached components.
- **Safety Tip:** Always press the plastic side-latch tab down fully before pulling to remove it, and never pull by the wires—pull on the plastic head.

2. SATA Power Connector



SATA power cable connector photo

- **Name:** SATA Power Connector
- **What it powers:** Hard drives (HDDs), solid-state drives (SSDs), and optical drives (like DVD drives).
- **Safety Tip:** Be gentle when plugging or unplugging; the plastic "L-shaped" guide inside can snap if inserted upside down or at an angle.

3. Molex Connector



Molex power connector cable photo

- **Name:** 4-Pin Molex Connector
- **What it powers:** Older hard drives, case fans, liquid cooling pumps, and accessories like RGB light strips.

- **Safety Tip:** Make sure the metal pins inside line up properly before pushing hard, as they can easily bend or push out of the plastic plug.

4. PCIe 6/8-Pin Connector



PCIe 8 pin power connector cable photo

- **Name:** PCIe 6-Pin / 8-Pin (6+2 Pin) Power Connector
- **What it powers:** Dedicated Graphics Cards (GPUs) that need extra power.
- **Safety Tip:** Ensure the connector is pushed all the way in until it clicks; a loose GPU power plug can overheat or melt during heavy gaming.

4.

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Install the Power Supply

1. Place the SMPS into the bottom of the PC case with its fan facing out (usually facing down).
2. Screw it securely into the back of the case using the provided screws.
3. Make sure the SMPS is **turned off** (the power switch on the back should be on the O position) and unplugged from the wall.

Step 2: Connect Power to the Motherboard

- **Cable used: 24-Pin ATX Cable and 4+4 Pin CPU Cable**
1. Take the large **24-pin connector** and line up its plastic clip with the matching socket on the right side of the motherboard. Push down firmly until it clicks.
 2. Take the **4+4 pin (CPU) connector** and plug it into the socket located at the top-left corner near the CPU.

Step 3: Connect Power to the Graphics Card (GPU)

- **Cable used: 6+2 Pin PCIe Cable**
1. Locate the PCIe power slots on the side or top of your graphics card.
 2. Push the 6-pin and 2-pin pieces together to form an **8-pin connector** (or use just 6 pins depending on your GPU).

3. Plug it firmly into the graphics card until the clip locks into place.

Step 4: Connect Power to Storage Devices (SSD / HDD)

- **Cable used: SATA Power Cable**

1. If you are using a 2.5-inch SSD or 3.5-inch Hard Drive, locate the wide "L-shaped" power port on the back of the drive.
2. Align the **SATA power connector** with the port and push it gently until it slides all the way in. *(Note: If you use an M.2 NVMe SSD attached directly to the motherboard, no extra cables are needed!)*

Step 5: Double-Check and Turn On

1. Double-check that all connectors are fully pushed in with no loose gaps.
2. Plug the main power cord into the back of the SMPS and wall outlet.
3. Flip the SMPS switch to I (On) and press your PC's power button!